WORKED SOLUTION

STEP II 1989, Question 3

Pure • Complex Hyperbolic Mapping

STEP II 1989, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q3

The key idea

The right-hand side is $2\sinh(x+\mathrm{i}y)$, so the whole question is about the map $u+\mathrm{i}v=\sinh(x+\mathrm{i}y)$. Expanding sinh of a sum splits it into a real and an imaginary part, giving u and v explicitly. Fixing x or y then leaves a parametric curve, and the identities $\cos^2+\sin^2=1$ and $\cosh^2-\sinh^2=1$ eliminate the parameter.

Solution

Finding u and v

Dividing by 2, $u+\mathrm{i}v=\sinh(x+\mathrm{i}y)$. Using $\sinh(\mathrm{i}y)=\mathrm{i}\sin y$ and $\cosh(\mathrm{i}y)=\cos y$,

$$\begin{aligned}\sinh(x+\mathrm{i}y) &= \sinh x\cosh(\mathrm{i}y)+\cosh x\sinh(\mathrm{i}y)\\ &= \sinh x\cos y+\mathrm{i}\cosh x\sin y.\end{aligned}$$

(If you prefer, expand $\frac{1}{2}(\mathrm{e}^x(\cos y+\mathrm{i}\sin y)-\mathrm{e}^{-x}(\cos y-\mathrm{i}\sin y))$ directly; it's the same thing.) Since x, y, u, v are all real, equating real and imaginary parts gives

$$u=\sinh x\cos y,\qquad v=\cosh x\sin y.$$

The line $x=a$

With $x=a$ fixed and y free, $u=\sinh a\cos y$ and $v=\cosh a\sin y$. Since $a>0$, both $\sinh a$ and $\cosh a$ are positive, so we may divide:

$$\left(\frac{u}{\sinh a}\right)^2+\left(\frac{v}{\cosh a}\right)^2=\cos^2 y+\sin^2 y=1.$$

Conversely every point of this ellipse has the form $(\sinh a\cos y,\cosh a\sin y)$ for some y, since that is the standard parametrisation of the ellipse. So the line $x=a$ corresponds to the whole ellipse (traced once for each interval of length 2π in y), with semi-axes $\sinh a$ along the u-axis and $\cosh a$ along the v-axis.

The line $y = b$

Now $y = b$ is fixed and x is free: $u = \sinh x \cos b$, $v = \cosh x \sin b$. We are told $0 < \sin b < 1$, so $\sin b \neq 0$ and $\cos^2 b = 1 - \sin^2 b > 0$, so $\cos b \neq 0$ too. Dividing,

$$\left(\frac{v}{\sin b}\right)^2 - \left(\frac{u}{\cos b}\right)^2 = \cosh^2 x - \sinh^2 x = 1.$$

This is a hyperbola with its axis along the v-axis and vertices at $(0, \pm \sin b)$.

Ask yourself Why only *one* branch, and is all of that branch covered?

Since $\cosh x > 0$ and $\sin b > 0$, we always have $v > 0$: only the upper branch is reached. And all of it is reached: given any point (u, v) on the upper branch, choose $x = \sinh^{-1}(u / \cos b)$, so that $\sinh x \cos b = u$; then $\cosh x \sin b$ is the positive root of the hyperbola's equation for that u, which is v. So the line $y = b$ corresponds to exactly one branch, the one with $v > 0$.

The intersection and the right angle

The point $x = a$, $y = b$ lies on both lines, so its image lies on both curves:

$$(u, v) = (\sinh a \cos b,\ \cosh a \sin b).$$

To find the directions of the curves there, differentiate each parametrisation with respect to its own parameter. Along the ellipse ($x = a$, parameter y), at $y = b$:

$$\mathbf{t}_1 = \left(\frac{\mathrm{d}u}{\mathrm{d}y}, \frac{\mathrm{d}v}{\mathrm{d}y}\right) = (-\sinh a \sin b,\ \cosh a \cos b).$$

Along the hyperbola ($y = b$, parameter x), at $x = a$:

$$\mathbf{t}_2 = \left(\frac{\mathrm{d}u}{\mathrm{d}x}, \frac{\mathrm{d}v}{\mathrm{d}x}\right) = (\cosh a \cos b,\ \sinh a \sin b).$$

Neither vector is zero, because each has a component $\cosh a \cos b \neq 0$, so both are genuine tangent vectors. Their scalar product is

$$\mathbf{t}_1 \cdot \mathbf{t}_2 = -\sinh a \sin b \cosh a \cos b + \cosh a \cos b \sinh a \sin b = 0,$$

so the ellipse and the hyperbola branch cross at right angles at this point, as required.

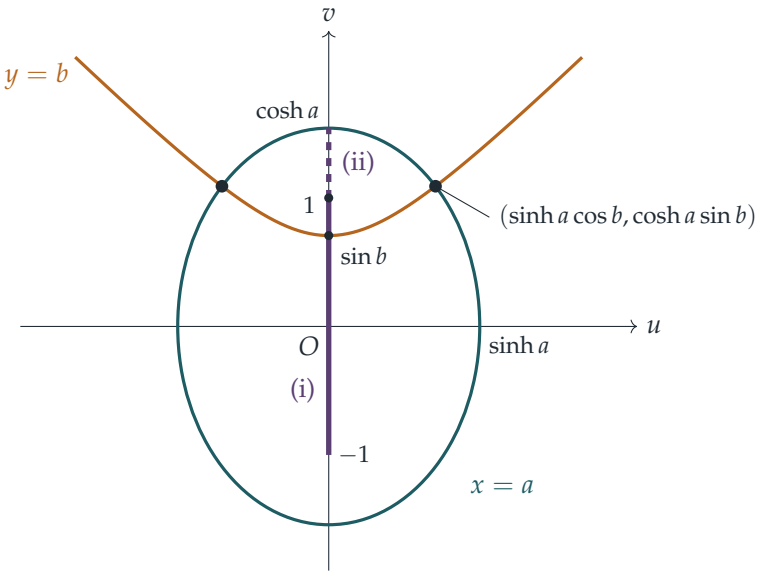
(By the symmetry $u \mapsto -u$ of both curves, they also meet at right angles at $(-\sinh a \cos b, \cosh a \sin b)$, the image of $x = -a$, $y = b$, or equally of $x = a$, $y = \pi - b$.)

The sketch

(i) $x = 0$ gives $u = 0$, $v = \sin y$: the segment of the v-axis from $(0, -1)$ to $(0, 1)$. Notice that it is the limit of the ellipses as $a \to 0$, since $\sinh a \to 0$ and $\cosh a \to 1$.

(ii) $y = \frac{1}{2}\pi$, $0 \leq x \leq a$ gives $u = 0$, $v = \cosh x$, which increases from 1 to $\cosh a$: the segment of the v-axis from $(0, 1)$ up to the top of the ellipse, $(0, \cosh a)$.

The hyperbola branch has its vertex at $(0, \sin b)$, which lies on segment (i). The sketch uses $a = 1$, $b = \frac{1}{4}\pi$.



Segment (i) is solid and segment (ii) dashed; together they run up the v-axis from -1 to the top of the ellipse. Taking $y = \frac{1}{2}\pi$ for all real x instead would give the ray $v \geq 1$ on the v-axis twice over, the degenerate case $\sin b = 1$ that the question excludes.

Checks

- Both conics have foci at $(0, \pm 1)$: for the ellipse $\cosh^2 a - \sinh^2 a = 1$, and for the hyperbola $\sin^2 b + \cos^2 b = 1$. Confocal ellipses and hyperbolas always cross at right angles, which agrees with our result.
- Numerically, with $a = 1$, $b = \frac{1}{4}\pi$: $\sinh(1 + \frac{1}{4}\pi\mathrm{i}) \approx 0.8310 + 1.0911\mathrm{i}$, matching $(\sinh 1 \cos \frac{1}{4}\pi,\ \cosh 1 \sin \frac{1}{4}\pi)$. We also checked the two curve equations and the zero scalar product at many random (a, b).

Common slips

- Writing $\sinh(\mathrm{i}y) = \sin y$ rather than $\mathrm{i} \sin y$, which swaps the roles of u and v.
- Saying "a hyperbola" without explaining why only the branch with $v > 0$ appears.
- Forgetting to check $\cos b \neq 0$ before dividing by it; that is exactly what the condition $\sin b < 1$ is for.

Beyond STEP

The map $z \mapsto \sinh z$ is *conformal*: wherever its derivative is non-zero it preserves angles, so the perpendicular grid of lines $x = \text{const}$ and $y = \text{const}$ must become a perpendicular grid of curves, here confocal ellipses and hyperbolas. This is why such "elliptic coordinates" appear when solving Laplace's equation around a flat plate or a slit. Needham's *Visual Complex Analysis* explains conformal maps beautifully, with many pictures of exactly this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.